Ysbyty Gwynedd
Anaesthetics Department
Adran Anaestheteg

PERIOPERATIVE

TIVA

YG CG 424 Mar 2024

Author(s): Dr Suzi Cary-Jones, Dr Stephan Clements
Reviewer: Departmental Guideline Group

FIRST APPROVED: 5/03/24 REVIEW BY: 5/3/2027

Key Points

- Individuals providing TIVA must be adequately trained and confident in the use of TIVA.
- NAP 5 showed that 47% of cases of AAGA where at induction, and 34% during maintenance, and 18% at emergence of anaesthesia.

Introduction

This guideline incorporates all 10 recommendations by the Association of Anaesthetists and relavant recommendations from NAP 5.

Abbreviations used in this document:
TIVA – Total Intravenous Anaesthesia
AAGA – Accidental awareness under General Anaesthesia
PONV – Post-operative nausea and vomiting

Considerations

Absolute indications:

Patients with a known or suspected diagnosis of Malignant hyperthermia requiring general anaesthesia.

Relative indications:

Patients requiring general anaesthesia who:

- Are high risk of PONV.
- Who are having prolonged surgery where a shorter emergence time would be beneficial
- Patients who have a family history of malignant hyperthermia, but have not been investigated themselves.
- Require head and neck surgery where delivery of anaesthesia may be more reliable via the IV route than the inhalational route.

Benefits of TIVA:

- Reduced risk of PONV compared to volatile anaesthesia
- Theoretical improvement in the risk of metastatic disease in patients with certain malignancies.
- Rapid emergence from anaesthesia with reduced emergence delirium.
- May reduce peri-operative pain and peri-operative neurocognitive disorders.
- May be given remotely where volatile anaesthesia is not available.
- In some head and neck surgery, may be more reliable and reduces exposure of theatre staff to volatile anaesthesia.
- Lower carbon-equivalent footprint than volatile anaesthesia.

Cautions:

- Access to peripheral cannula difficult intra-operatively.

Risks of TIVA:

- Pain on injection – common
- Increased risk of AAGA: (1/19,000 for all GAs). Discuss this risk with the patient and explain what the patient may experience until a suitable level of anaesthesia is reached, particularly when planning to carefully titrate to effect eg. in the elderly.
- Opioid induced hyperalgesia, if high doses of remifentanil are administered.
- Cardiovascular instability in frail patients when the dose of TIVA is not carefully titrated.

Equipment & Drugs

Equipment:

- Use dedicated TCI pumps only.
- Only use dedicated TIVA lines which have the following key features: minimal deadspace, Luer-lock connector ends, antisyphon valves and an anti-reflux valve on fluid administration line.

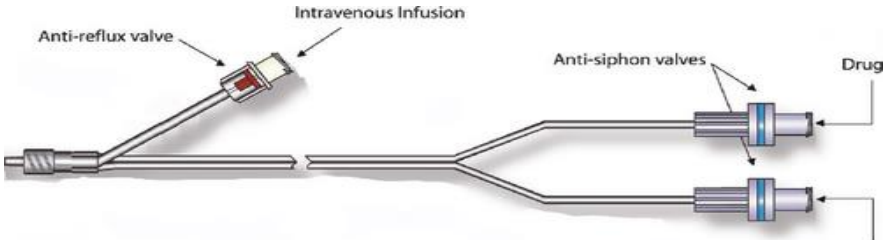


Fig.1 multi-lumen connector including an anti-reflux valve for intravenous fluid and anti-siphon valves for intravenous drugs. (AA Guidelines,2019.)

- **Bispectral index (BIS) monitoring:**
  - BIS monitoring is mandatory. Check it is working prior to induction of anaesthesia.
  - Target BIS 40-60. Assess adequate depth prior to intubation.
  - In cases where patients are transferred from a volatile anaesthesia to IV anaesthesia at the end of an operative case e.g. for transfer to ITU or for CT scan, BIS must be used prior to conversion to minimise risk of awareness. *(NAP5 18.3)*

Drug concentrations:

- Always use 1% propofol for TIVA.
- Remifentanil must be drawn up at a concentration of 50mcg/ml. (Usually 2mg in 40mls of saline 0.9%).
- Drug concentrations should be double checked at the time of drawing up and when programming the pumps.

Appropriate cannula sites:

- Ideally, readily accessible for checking & flowing well.
- Have a low threshold for siting a second, back up cannula when access to the patient is limited e.g. robotic cases and dynamic hip screws. *(NAP5 18.2)*

Models

- The Eleveld model for Propofol & Remifentanil is the preferred model, being derived from the largest dataset that spans the widest age and body weight range, thereby minimising prediction errors.
- The following graphs are a useful starting point for Propofol Eleveld. As the induction bolus is larger than older models, care should be taken with fragile patients.
- Set the initial target to the Ce50 value from the right hand graph, then later titrate to effect using the left hand graph as a guide.

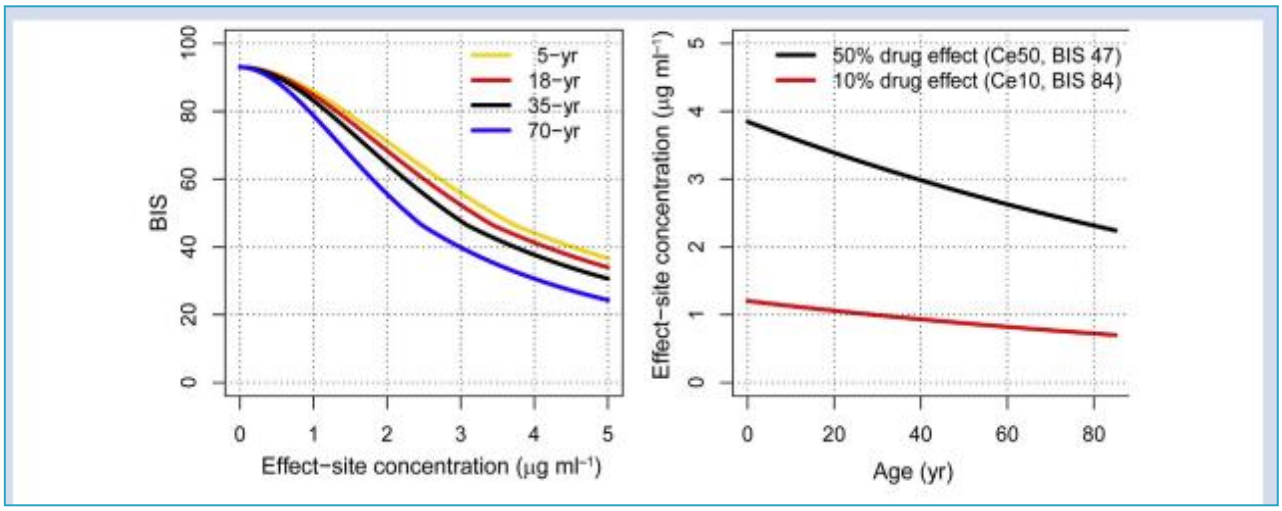
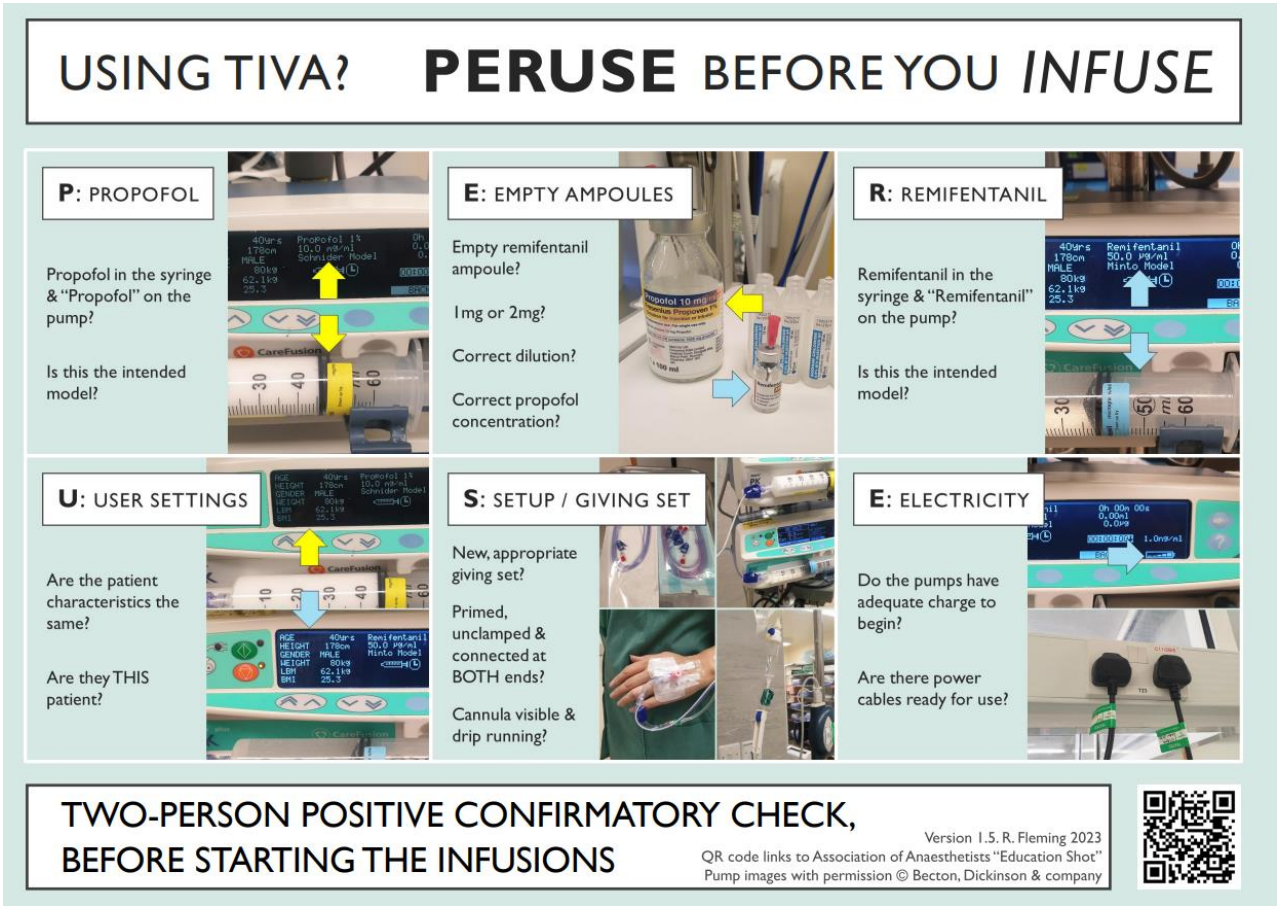


Fig.2 Propofol Eleveld Model: Relationships between effect-site concentrations & predicted BIS & the relationship between age & effect-site concentration for 50% & 10% drug effect.

- The starting infusion boluses and rates must be carefully considered, especially in frail, elderly patients. In these cases it is often safest to start low and slowly titrate up, having vasopressors to hand, and ensuring optimum cardiovascular filling status pre-induction.

Infusion Pump Setup



- Load the syringe containing the correct drug first, then programme the pump.
- Keep distractions to a minimum.

Intra-Operative Monitoring

- Extra-vigilance should be paid to depth of anaesthesia immediately after intubation when monitoring may be less visible for transfer at the same time that the pumps may pause temporarily as part of their modelling programme.
- Check the TIVA iv line regularly & document. When the cannula is inaccessible, the high pressure alarms must not be altered, as this may delay recognition of failed drug delivery.
- Record BIS level every 5 minutes & IV line check every 15 minutes.

A Pre-Knife to Skin Checklist:

- BIS displayed and appropriate?
- HR/BP consistent with adequate anaesthesia
- Cannula patency checked? Back-up if loss?
- TIVA pumps running at appropriate dose and battery level adequate?

- Discuss with ODP what steps will be taken if a) possible AAGA, b) a pump failure occurs.

Neuromuscular Reversal:

- Ensure a train-of-four ratio of >90% prior to allowing the BIS to rise to awake levels. Ideally the BIS should remain at <60 prior to full neuromuscular block reversal. Do not remove BIS monitoring until full neuromuscular block reversal is achieved. *(NAP5 10.6)*

Emergency Checklists

? Depth of Anaesthesia Inadequate:

- Ask the surgeons to pause what they are doing (rarely this will be inappropriate).
- Check for signs of awareness e.g. rising BIS, tachycardia, hypertension, tearing, any movement.
- Check cannula site? Tissued? Consider switching to a different cannula.
- Consider increasing propofol infusion rate.
- Consider switching to sevoflurane.
- Do not allow the surgeons to recommence surgery, or give further muscle paralysis until you are happy that depth of anaesthesia is adequate.
- If you think awareness under anaesthesia is likely, speak to the patient calmly and let them know that you are doing everything possible to resolve the situation.
- Document all of your actions clearly.

Pump Failure:

- This is a time critical emergency. Have a low threshold for calling for help!
- Is conversion to inhalation contraindicated (e.g. MH)?
    - Yes: Call for help, bolus propofol while re-programming a pump.
    - No: Turn on volatile while you figure out what next.
- Options:
    - A: Continue with volatile anaesthesia.
    - B: Re-program a new pump:
        - Program as per previous pump. Double check! You may be stressed!
        - Before attaching to patient, press start & discard the initial bolus, then attach to patient. Careful titration to the BIS and patient assessment will be required. 'Bridging' boluses of propofol during set-up may also be required.
        - N.B. Displayed target concentrations will not be accurate.

Possible AAGA:

- Discuss with the Consultant responsible for the list.
- Check on the patient in recovery.
- Download & use NAP5 support pack (Link below).
- Complete a DATIX
- Further investigation and support for the patient may be required.

Training

Individuals providing TIVA must be adequately trained and confident in the use of TIVA. If another colleague provides care during part of the anaesthetic, they must be happy with the model in use, and be independently competent at managing potential issues with TIVA intra-operatively.

References:

- https://bjanaesthesia.org/article/S0007-0912(18)30051-5/fulltext
- Nimmo et al, Guidelines for the safe practice of total intravenous anaesthesia (TIVA)
- **Joint Guidelines from the Association of Anaesthetists and the Society for Intravenous Anaesthesia. Oct 2019. Association of Anaesthetists. https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.14428**

Resources:

- The 'NAP5 Handbook': Concise practice guidance on the prevention and management of accidental awareness during general anaesthesia:

https://anaesthetists.org/LinkClick.aspx?fileticket=IrKEe85PVQQ%3d&portalid=0

- NAP5 Anaesthesia Awareness Support Pack

https://www.nationalauditprojects.org.uk/NAP5-Anaesthetia-Awareness-Pathway#pt